Ex. No: 13

Title: Sequential Dataset Preparation and RNN Prediction

Packages: TensorFlow, Keras, NumPy, Pandas, Matplotlib

AIM:

To prepare a sequential dataset and predict the next value of the sequence using a Recurrent Neural Network (RNN).

ALGORITHM:

1. Start the program and import the required libraries.
2. Create a sine wave and store it in a Pandas DataFrame.
3. Prepare the sequences: use 10 previous values as input and the next value as output.
4. Reshape the data to (samples, time steps, features) and split it into training and testing sets.
5. Build the model with a SimpleRNN layer and a Dense output layer.
6. Compile the model using the Adam optimizer and MSE loss, and train it for 20 epochs.
7. Predict on the test data and print the Mean Squared Error.
8. Plot the actual and predicted values and stop.

PROGRAM:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tensorflow import keras
from tensorflow.keras import layers

# Sequential data: sine wave stored in a DataFrame
t = np.arange(0, 100, 0.1)
df = pd.DataFrame({"time": t, "value": np.sin(t)})
print(df.head())

# Prepare sequences: 10 previous values -> next value
window = 10
values = df["value"].values
X, y = [], []
for i in range(len(values) - window):
    X.append(values[i:i + window])
    y.append(values[i + window])
X = np.array(X).reshape(-1, window, 1)
y = np.array(y)

split = int(0.8 * len(X))
X_train, X_test = X[:split], X[split:]
y_train, y_test = y[:split], y[split:]

# RNN model
model = keras.Sequential([
    layers.Input(shape=(window, 1)),
    layers.SimpleRNN(16),
    layers.Dense(1)
])
model.compile(optimizer="adam", loss="mse")
```

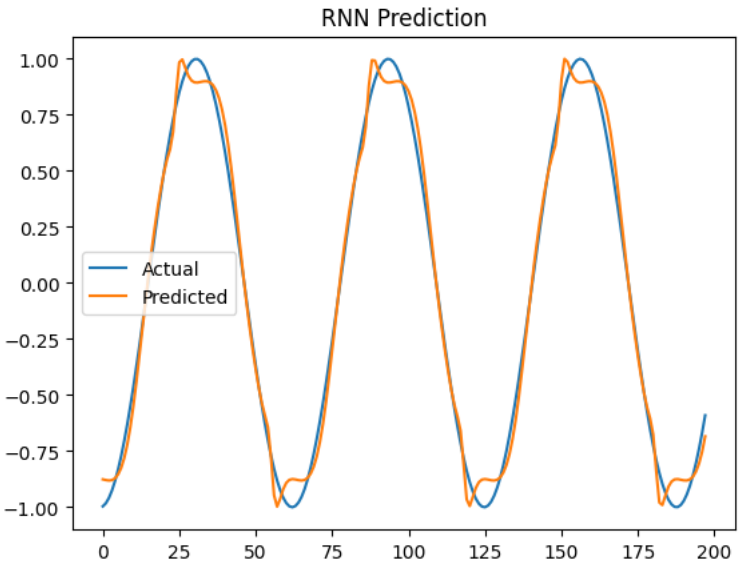
```
model.fit(X_train, y_train, epochs=20, verbose=0)

print("Test MSE =", round(model.evaluate(X_test, y_test, verbose=0), 5))

pred = model.predict(X_test, verbose=0)
plt.plot(y_test[:200], label="Actual")
plt.plot(pred[:200], label="Predicted")
plt.title("RNN Prediction")
plt.legend()
plt.show()
```

OUTPUT:

```
...     time     value
    0   0.0  0.000000
    1   0.1  0.099833
    2   0.2  0.198669
    3   0.3  0.295520
    4   0.4  0.389418
    Test MSE = 0.00493
```



RESULT:

The sequential dataset was prepared and the RNN predicted the next values of the sequence successfully with a very low error.